

P. Brown

Eighth Annual In-House Convention

26 & 27 April 1982

Room 4258/63

Department of Psychology

Dalhousie University

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Schedule

MONDAY, APRIL 26

2:00 - 3:15

Session I. Chair: R. Klein

- R. Rodger - Welcome
- ①. R. Brown
- 2. B. Fantie
- 3. A. Warner
- 4. S. Thorpe

COFFEE BREAK

3:45 - 5:00

Session II. Chair: L. Smith

- 5. C. Shaw & M. Cynader
- 6. S. Bryson
- 7. K. Hadskis, T. Minor,
H. Robertson & V. LoLordo
- 8. J. Radel

5:00

BEER IN LOUNGE

TUESDAY, APRIL 27

9:30 - 10:30

Session III. Chair: D. Henken

- 9. L. Smith & R. Klein
- 10. D. Mitchell, M. Ruck & M. Kaye
- 11. J. Mendelson
- 12. J. Fairless

COFFEE BREAK

11:00 - 12:00

Session IV. Chair: L. White

- 13. W. Honig & P. Dodd
- 14. P. McLeod
- 15. K. Grasse
- 16. S. Shaw

12:00 - 2:00

CATERED LUNCH & POSTER SESSION

- 17. R. Klein & T. Kelbrat
- 18. D. Clark

2:15 - 3:15

Session V. Chair: C. Reed-Elder

- 19. I. Meinertzhagen, R. Menzel
& G. Kahle
- 20. J. Earle
- 21. M. Kaye, D. Mitchell
& M. Cynader
- 22. B. Rusak

COFFEE BREAK

3:45 - 4:45

Session IV. Chair: R. Rodger

- 23. M. Cynader, F. Leporé &
J.P. Guillemot
- 24. B. Fantie
- 25. T. Minor & V. LoLordo
- 26. J. Fentress

4:45

BEER & PIZZA IN THE LOUNGE

ABSTRACTS

1. R. Brown

The Development of Odour Preferences in 18 to 20 Day Old Rats.

Rats were reared with their mother alone, their mother and father together or 12 hrs per day with their mothers and 12 hrs per day with their fathers separately. When tested for odour preferences with a set of "female odours" [own mother (OM), alien mother (AM), non-lactating female (NLF), and no-odour control (NOC)], both males and females showed the preference scale OM>AM>NLF>NOC, regardless of rearing condition. When tested with a set of "male" odours (own father, alien father, non-father male, and a no-odour control), male odours were preferred to the no-odour control, but no male odour was preferred to another male odour. These results indicate that the odour of the lactating female is preferred to other odours which rats experience during the same period. These results also suggest that infant rats are able to discriminate between their own mother's odour and that of another lactating female.

2. B. Fantie

Septal Modulation and Long-Term Potentiation in the Hippocampus:
A Neuronal Model of "Physiological Context"?

Stimulating the entorhinal cortex produced a monosynaptic evoked potential in the ipsilateral hippocampus. High frequency stimulation causes long periods of marked increase in response to a test stimulus (Long Term Potentiation, LTP). Pairing entorhinal and septal stimulation augments the population spike component of this evoked potential. Simultaneous high frequency stimulation of entorhinal cortex and septum enhances LTP and septal augmentation only in the hemisphere in which septal and entorhinal stimulation were synchronous.

3. A. Warner

Maximizing, Therapeutic Benefits and Minimizing Professional Involvement.

The number of parents seeking assistance and guidance in coping with their children's behavior developments and problems is much greater than the number of mental health professionals available to work with them. One solution is the development of videotape programs on child development issues for parents which provide important information and coping strategies. They can reach larger numbers of people with interesting, yet conceptually sound information. The presentation primarily consists of the screening of one such videotape, part one of a three part series on introducing parents to information, methods, and potential problems in toilet training.

4. S. Thorpe

Hebb's Cell Assembly Theory Revisited.

5. C. Shaw and M. Cynader

Postsynaptic Control of Cortical Plasticity.

Various models of cortical plasticity are divided about the role that the cortical cell actually plays when the brain is altered as a result of unusual visual experience. We set out to examine the role of the postsynaptic cell activity in this plasticity process by constantly infusing L-glutamate onto the cortex during a plasticity paradigm, allowing the other side of the brain to serve as a control. L-glutamate causes uncontrolled firing of cortical cells and breaks down the correlation between environmental input and cellular response. Our results indicate that this breakdown of input/output correlation prevents environmental modification of the visual cortex.

6. S. Bryson

Hemispheric Asymmetries for Letters and Their Mirror Images.

A series of reaction time studies were conducted to determine how memory for the lateral orientation of letters might be mediated by the human brain. Single letters were presented to each visual field. The letters appeared in both their forward and backward orientations. Right-handed adults used a manual response to distinguish between 1) two letters, 2) upper- and lowercase letters, and 3) letters and numerical characters. In each experiment, forward letters were identified faster than backward ones when they were presented to the right visual field, whereas no forward-backward letter difference was found for the left visual field. The findings are discussed in terms of the idea that memory for the correct orientation of letters is generally lateralized to the left hemisphere.

7. K. Hadschis, T. Minor, H. Robertson and V.M. LoLordo

Signals for Safety and Benzominergic Receptors.

Diazepam produces muscle relaxation and relief from anxiety by occupying benzodiazepine receptors located throughout the central nervous system. Stimuli predicting safety in an otherwise dangerous situation also produce relief and relaxation. To determine whether safety signals exert their influence by releasing an endogenous benzodiazepine, rats were exposed to a situation in which a safety period was either present or absent. Despite large differences in behavior, ulcerations, and plasma corticosterone no differences were found for binding of H^3 diazepam.

8. J. Radel

Degeneration of Optic Nerve Fibers in the Goldfish Tectum.

I will present a brief account of the ultrastructural changes occurring among the optic fibers within the tectum at different times following optic nerve crush.

9. L. Smith and R. Klein

Attention, Automaticity, and the Cerebral Hemispheres.

Can one hemisphere become skilled at a task, while the other remains a novice? Each hemisphere learns and practices separate attention-demanding tasks, and when performance for both tasks/hemispheres becomes effortless (automatic), the task is changed for one but not the other hemisphere, or each hemisphere is tested on the task learned by the other one. The pattern of performance, compared with the 'whole brain' case, will have relevance to the nature of interhemispheric communication, attentional resources, and the mechanism of automaticity.

10. D. Mitchell, M. Ruck and M. Kaye

Short-term Effects on Visual Acuity of Surgically Induced Strabismus in Kittens.

In order to throw more light on the origin of strabismic amblyopia, frequent measurements were made of the visual acuity of each eye of young kittens following imposition of either a surgically-induced esotropia or exotropia. Whereas the immediate effects of the surgery were quite substantial in the case of kittens rendered esotropic (cross-eyed), the effects on the exotropic kittens were slight and quite transitory. These differences in the immediate effect of surgery may provide a partial explanation for the vast differences in the degree of amblyopia exhibited eventually by the two groups of strabismic kittens.

11. J. Mendelson

Directional Selectivity in the Cat Primary Auditory Cortex.

Extracellular responses to frequency modulated sweeps were examined in the cat primary auditory cortex. The stimuli presented were; 1) in-phase binaural sweeps, 2) anti-phase binaural sweeps, and 3) monaural sweeps (in both directions) to the contralateral only and to the ipsilateral ear only. Cellular responses were used to initially group cells as EE or EI. Cells were then classified according to the preferred direction of the sweep. Preliminary results showed that of the EE cells, 43% preferred up sweeps in both ears, 14% preferred down sweeps in both ears, and 43% exhibited no directional preference. Of the EI cells, 50% preferred an up sweep to the contralateral ear and a down sweep to the ipsilateral ear, 33% preferred a down sweep to the contralateral ear and an up sweep to the ipsilateral ear, and 18% exhibited no directional preference. The results imply the EE cells may be involved in localizing sounds which are presented along the vertical axis while EI cells may be used for localizing sounds along the azimuthal axis.

EE Binaural Amblyopia
EI Binaural Amblyopia

12. J. Fairless

Context Effects in the Approach-Withdrawal Behavior of Pigeons.

When an illuminated key is presented repeatedly in some relation to food, the keylight eventually gains considerable control over the behavior of hungry pigeons. The form of this acquired responding is dependent on the relative rate of reinforcement during and in the absence of key illumination. Approach to the keylight develops when its presentation is associated with a food rate greater than that associated with the overall situation, while withdrawal from the keylight develops when its associated food rate is less than that otherwise available. Current research in our lab indicates that traditional characterizations of the learning which underlies this behavior may be too narrow. Our data suggest that approach-withdrawal behavior reflects a primitive decision making process in which reinforcement rates associated with various stimuli are compared prior to the production of a response.

13. W.K. Honig and P.W.D. Dodd

Retention Interval Duration as a Discriminative Cue for Pigeons.

Responding to a vertical line on the pigeon's key yielded reinforcement if that line followed a 2-sec retention interval (RI), while horizontal was correct following a 10-sec RI. In both cases a white light marked the onset of the RI. Pigeons learned this discrimination. When the white light was replaced by red and green, correlated with RI duration, it was found that the new initial colors acquired little stimulus control. Thus, duration of the RI can serve as a salient cue which can block or overshadow control by more traditional visual cues.

14. P. McLeod

Vocalizations of the Pilot Whale (*Globicephala melaena*).

Vocalizations produced by wild pilot whales were recorded in the inshore waters of Newfoundland. Sounds were categorized on the basis of acoustic properties and temporal associations to create a catalog of the species vocal repertoire. One sample of 43 minutes duration (1408 phonations) was used to look for sequential patterning of sounds.

Underwater broadcasts of sounds to pilot whales were conducted. No consistent reaction to broadcasts was found.

A comparison of the vocalizations recorded from different pods showed considerable differences in the sounds produced. The importance of measuring variability and the possible sources of variance in the vocalizations of social delphinids are discussed.

15. K. Grasse

Self-Motion, Gibsonian Flow-Fields, and the Accessory Optic System.

16. S.R. Shaw

Novel, Inhibitory, Adaptive Control of Photoreceptor Output.

A main function of the first synapse in the visual pathway is to match the system operating range to the ambient lighting level, which can vary enormously, $>10^6$. In insects, the main control point is the photoreceptor terminal itself, and control is exercised by novel mechanisms of presynaptic lateral- and self-inhibition. Our results using intracellular recording and EM mapping techniques suggest that the several processes of one particular cell must be acting independently of one another. If this is confirmed, it will provide the first clear evidence contradicting the century old 'Neurone Doctrine' of Cajal.

17. R. Klein and T. Kelbrat

Hemispheric Asymmetries in Global and Local Processing.

Navon has proposed that the extraction of global information in a visual stimulus precedes and directs the extraction of local features. Our reading of the recent global/local literature suggests instead the parallel operation of two isolable subsystems. The purpose of our experiment was to test the twin hypotheses that the left hemisphere is the local processing system and the right hemisphere is the global processing system.

18. D. Clarke

Tooling Up For Clinical Auditory Neurophysiology.

Clinical research has been underway for several years now in labs around the world using multiple channel electrical stimulation of the auditory nerve ("cochlear implants") as a means of alleviating inner ear deafness (a problem afflicting some 200,000 North Americans). Research in this lab, using the domestic cat, will continue the auditory nervous system single unit studies which have made a major contribution to this advance. The requisite lab facilities are almost all in place with special attention having been given to stimulus presentation. With the aid of specialists in acoustics, sound delivery and measurement systems have been designed that are hoped to make an innovative contribution to the field.

19. I.A. Meinertzhagen, R. Menzel and G. Kahle

Spectrally Encoded Pathways in the Dragonfly Visual System.

Much circumstantial evidence suggests that dragonflies have highly developed colour vision, different species having different wavelength sensitivities. Intracellular recordings followed by injections of the dye lucifer yellow enabled the localisation of receptors with identified spectral sensitivity within one species, *Sympetrum rubicundulum*. Combined with receptor correctivity analyses (Meinertzhagen and Armett-Kibel) the results suggest that broad band green-sensitive receptors provide the predominant input to two visual interneurons, while orange-, UV- and violet-sensitive receptors provide selective inputs to two other interneurons.

20. J. Earle

Analgesia and Preference for Signalled Versus Unsignalled Shock.

Rats will spend more time in the half of a shuttlebox in which shocks are always predicted by a tone than on the other side, where shock is unsignalled. One account says that signals that predict shock acquire the capacity to elicit analgesia and attenuate the aversiveness of shock. Therefore the rat prefers the side on which the shocks hurt less.

We have shown that analgesia is elicited by a signal for shock. In rats which receive a tone randomly with respect to shock neither preference nor analgesia results.

The analgesia in the paired groups was found to be unaffected by injections of opiate antagonists, suggesting that the mechanism underlying the analgesia was not endorphinergic.

21. M. Kaye, D. Mitchell and M. Cynader

Consequences for Binocular Depth Perception of Neonatal and Adult Lesions of Cat Visual Cortex.

We have shown previously that removal of cortical areas 17 and 18 in adult cats produces a moderate reduction in visual acuity and severe deficits in depth perception, apparently including a loss of stereopsis. We report here that removal of these areas during the first month of life permits development of normal acuity and monocular depth discrimination, but the total absence of binocular advantage suggests a selective elimination of stereopsis.

22. B. Rusak

Physiology of Visual Projections Influencing Circadian Rhythm Entrainment.

I have previously used behavioural, lesion and stimulation techniques to study the neural mechanisms regulating mammalian circadian rhythms. I am now beginning neurophysiological studies of the visual pathways that may be involved in rhythm entrainment. I will describe my preliminary attempts to record in visual structures related to the circadian system and my plans for future work.

23. M. Cynader, F. Leporé, and J.P. Guillemot

Interhemispheric Communication and Postnatal Development.

The corpus callosum is a bidirectional pathway interconnecting the two cortical hemispheres. We have been able to influence the functional effectiveness of the part of the callosal pathway dealing with the visual cortex by allowing animals restricted visual exposure. We have arranged the right hemisphere to receive information only via a pathway through the left hemisphere throughout postnatal development. This results in a change in the functional capabilities of the corpus callosum so that it becomes a one-way pathway rather than a two-way path.

24. B. Fantie

The Effect of Constant Light on Sexual Behaviour of Virgin Male Rats.

After 6 months of exposure to constant light (LL) only 2 of 16 male rats ejaculated during their initial exposure to a receptive, soliciting female compared with 16 of the 18 maintained on a light/dark (12:12) schedule. Six of the LL rats were then switched to a LD schedule for 5 weeks while the remaining 6 were kept on LL. When tested again, 4 LD and 1 LL rat ejaculated.

25. T. Minor and V. LoLordo

Escape Deficits Following Inescapable Shock in the Rat: The Role of Odor and a Submissive Action Pattern Hypothesis.

Rats were exposed to escapable, inescapable, or no electric shock in small wheel-turn chambers containing odors of previously shocked rats. All rats were tested for the acquisition of lever-press escape 24 hr later in the presence of a novel odor, or the odor of previously shocked rats. The characteristic deficit in escape performance for inescapably shocked rats was absent when testing occurred in the presence of the novel odor. These data suggest that learned helplessness is transferred via odors in the rat.

26. J. Fentress

Stretching the Striatum.

The striatum is part of the mammalian brain that has long been thought to play an important role in integrated action. In this talk I shall briefly review some findings on striatal development obtained during my sabbatical at The Salk Institute, and offer some suggestions about how the role of the striatum might be assayed at several levels of behavioral expression.